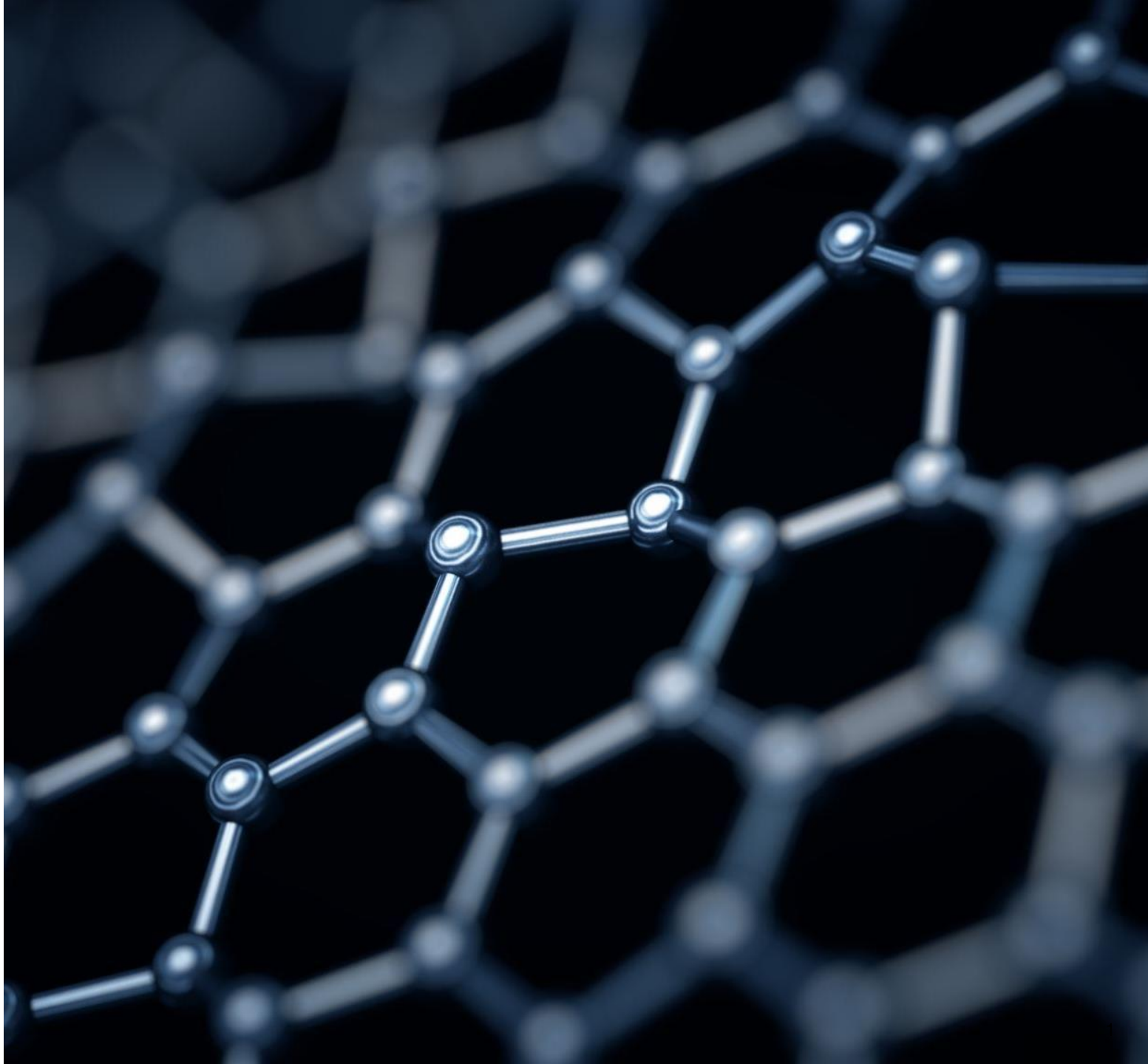
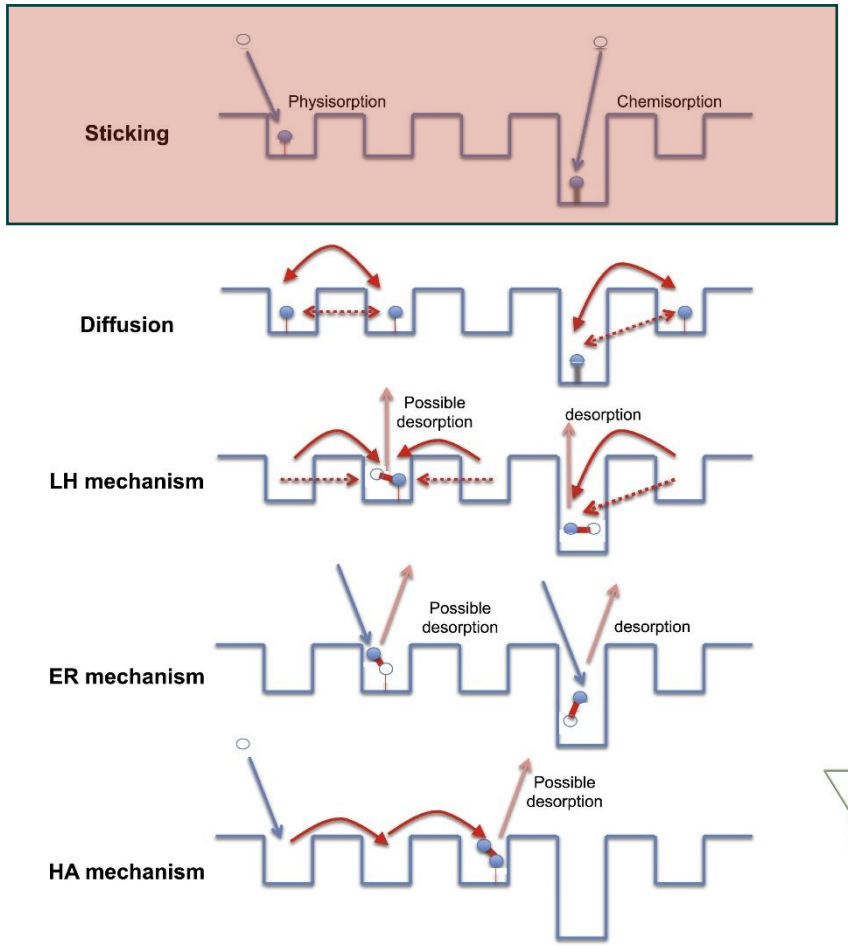
HYDROGEN STICKING DYNAMICS ON GRAPHENE SURFACE

Mukul Dhiman

Sabine Morrisset (ISMO)

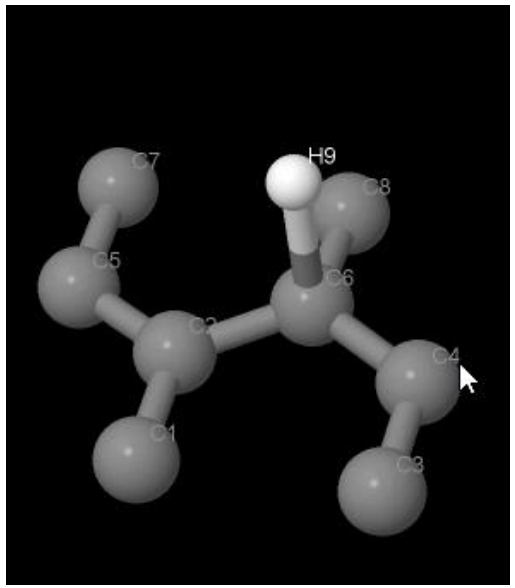


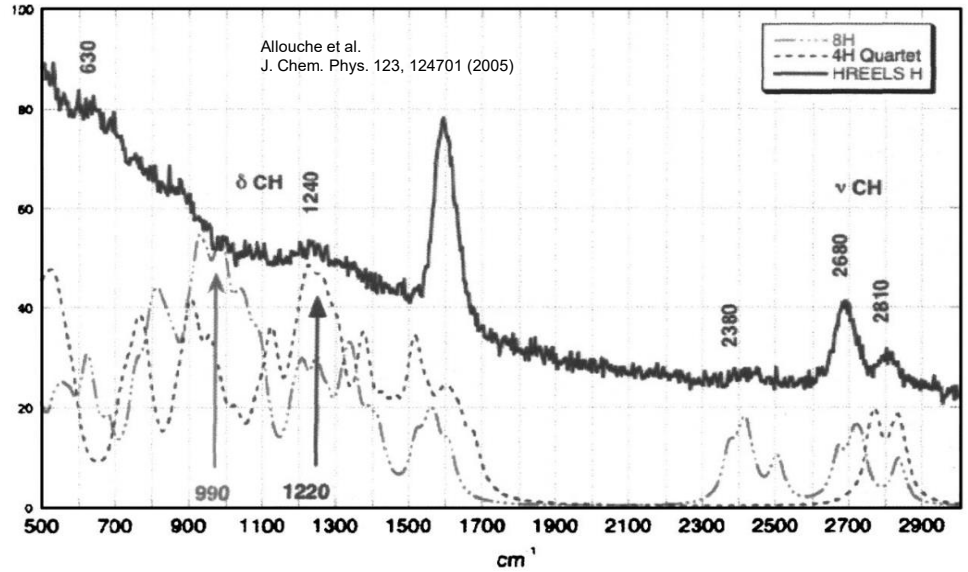
A brief recap of Hydrogen on Graphene

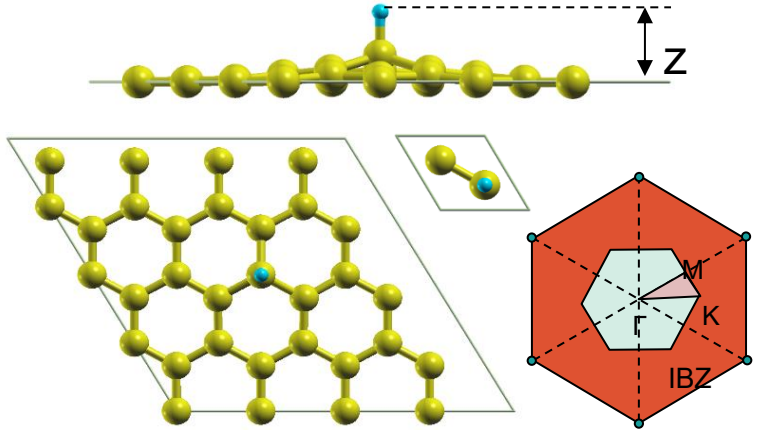


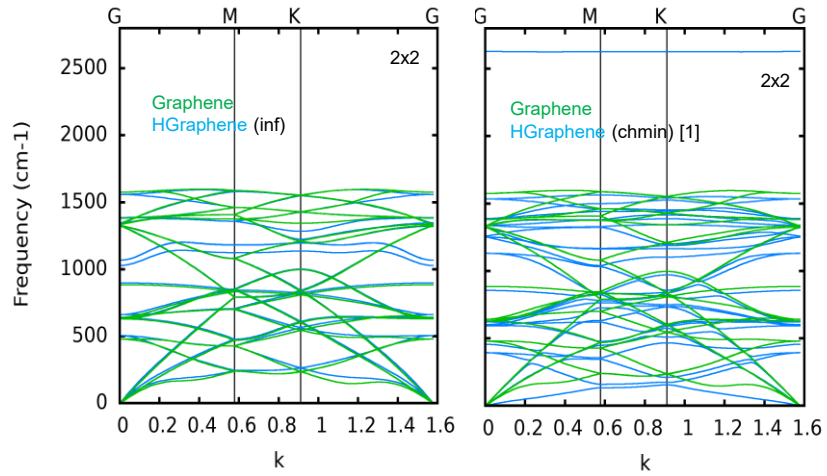
Molecular Hydrogen in ISM

V. Wakelam et al. / Molecular Astrophysics 9 (2017)









Sticking Dynamics

H Sticking Dynamics: Understanding the sticking of H on graphene is important for low energy interstellar medium where a physiosorbed H would be chemisorbed through **quantum tunnelling**.

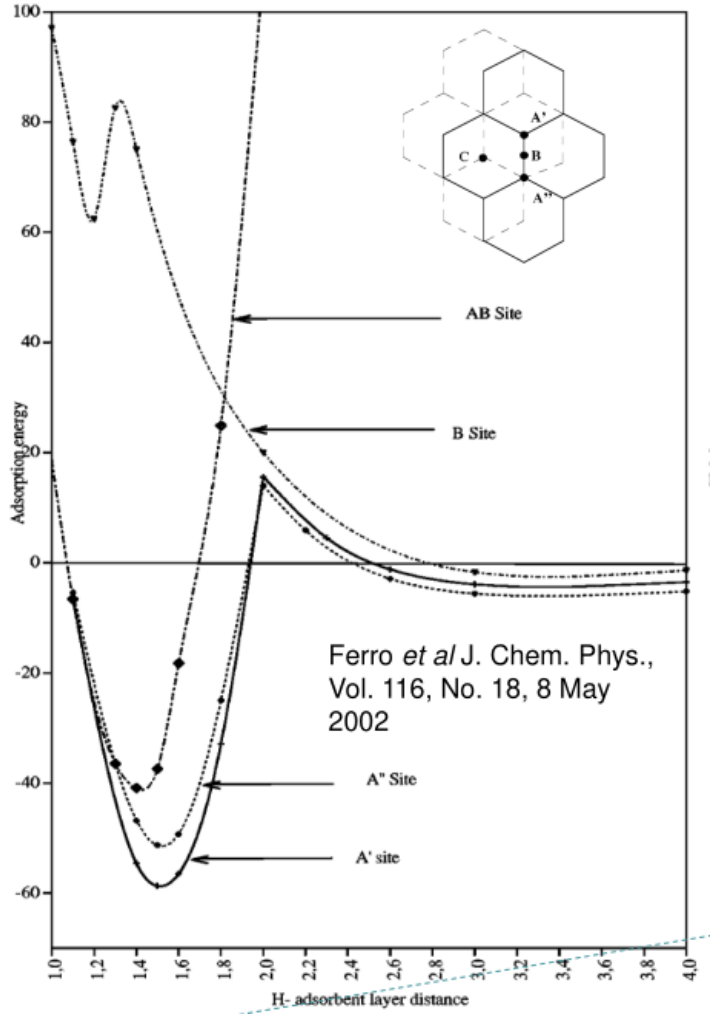
- **Quantum Methods** (with T)
  - Morisset *et al.* [1] (DFT+CCWP method)
  - Karlicky *et al.* [2] (**Chemisorption** study using system bath model)
  - Lepetit *et al.* [3,4] (**Physisorption** study with several methods)
- **Mixed Quantum Classical** (with T)
  - Morisset *et al.* [5] (Classical Phonons + H motion 1D WP)
- **MCTDH**: Bonfanti *et al.* [6,7,8] (T=0K)

- **Objective:**
  1. Single dynamics representing sticking probabilities in physorption well and chemisorption.

  2. Tunnelling mediated sticking at collision energy < 0.2eV with surface temperature between 10K to 100K. Experiment by Zecho *et al.* [9] on graphite and Haberer *et al.* [10] on Graphene observe this at 150K-300K.

  3. To study the sticking of H atom on graphenic surfaces using a new dynamics method developed in the **ANR HyTRAJ** (a hybrid quantum trajectories method) including phonon in dynamics.

Potential Energy Surface



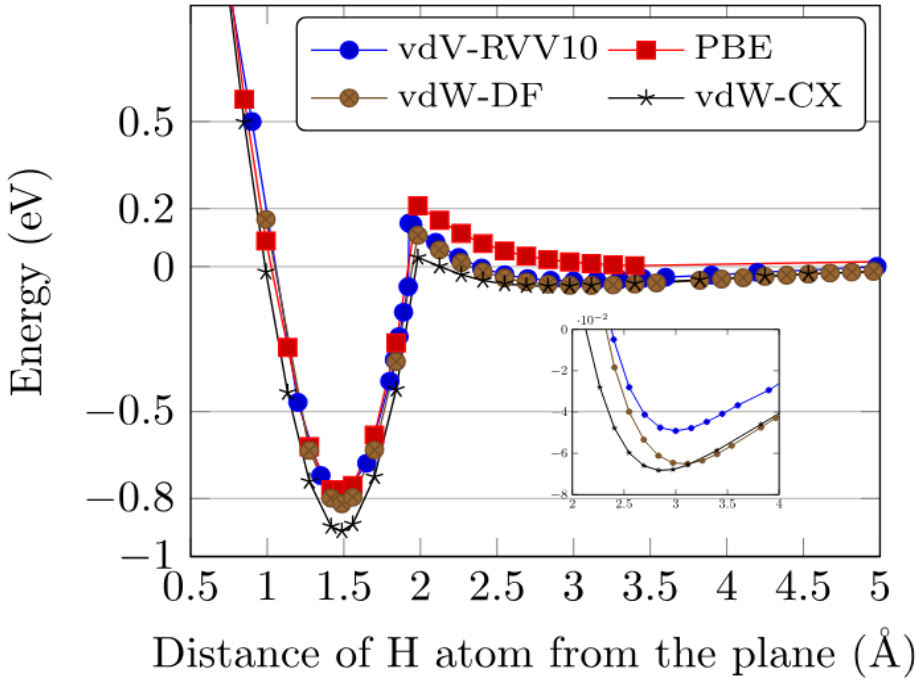


Table 1
Barrier Height ($E_{barrier}$), Chemisorption Energy (E_{chem}) and Physisorption Energy (E_{phys}) in eV as calculated using DFT in Quantum Espresso 7.2 for a single Hydrogen atom on Graphene surface.

Functional	$E_{barrier}$ (eV)		E_{chem} (eV)		E_{phys} (eV)	
	This work	Lit	This work	Lit	This work	Lit.
PW91 [37]	–	0.185 [12]	–	−0.608 [12]	–	−0.031[12]
PBE [38]	**0.188**	0.3 [8]	**−0.784**	−0.8 [8]	No	No [8]
vdW-RVV10 [33]	**0.169**	0.179 [8]	**−0.765**	−0.851 [8]	**−0.0294**	−0.029 [8]
vdW-DF-CX [32]	**0.031**	–	**−0.905**	–	**−0.0677**	–
vdW-DF [30,31]	**0.107**	–	**−0.812**	−1.10 [36]	**−0.0645**	−0.020 [36]
MP2	–	–	–	–	–	−0.0345 [10]

++ Ivanovskaya et al. / Rougeau et al. / Wang et al.

rvv10-> non-local vdW-DF

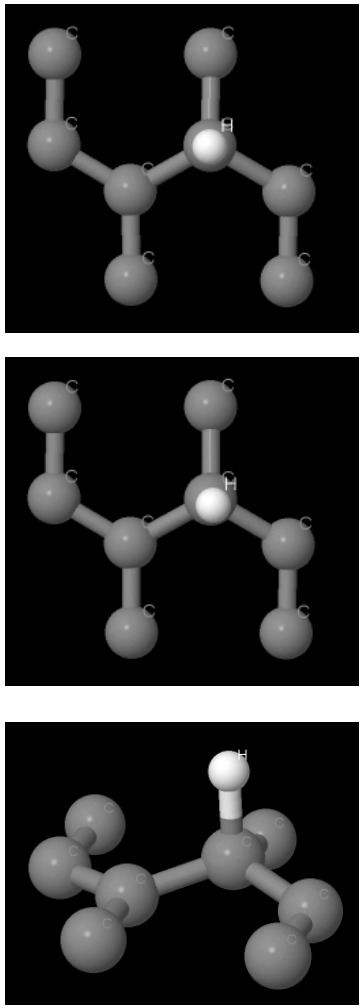
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Phonon Calculations

Surface modes for sticking

Γ- point calculation for chemisorbed H atom

number	vdW-DF	vdW-DF-CX	vdW-RVV10	PBE
1	-864.900109	-774.443572	-155.843572	49.213912
2	-864.900109	-774.443537	-102.064976	49.213912
3	-745.741165	-616.43166	-102.064828	103.898705
4	-676.973448	-275.761449	343.953335	395.159297
5	-644.335961	-275.760561	343.953623	395.170684
6	-644.335961	168.061103	449.67022	461.286595
7	-439.485144	418.206046	564.917395	590.724811
8	-439.485144	418.206102	564.917478	590.728089
9	416.831269	470.125596	591.6489	601.297983
10	464.284689	629.425177	611.885963	621.656535
11	464.284689	768.41307	612.120676	621.663507
12	541.173071	768.413107	612.120716	626.196165
13	674.070333	799.12411	839.760371	855.185764
14	788.313219	966.88121	1125.08559	1127.003002
15	931.142905	966.88128	1125.085684	1127.011600
16	931.142905	989.872528	1239.859871	1248.649052
17	1147.185712	1154.133163	1239.859997	1248.672517
18	1196.487048	1222.888953	1263.246687	1259.547605
19	1196.487048	1222.889074	1308.575435	1321.394910
20	1211.813237	1226.919149	1321.336328	1336.777999
21	1211.813237	1226.919561	1337.235985	1336.782407
22	1261.64491	1265.714862	1337.236255	1339.981238
23	1328.227509	1333.27116	1376.559323	1388.931094
24	1328.227509	1333.271409	1376.559731	1388.940513
25	1394.508292	1403.712742	1524.152361	1534.732433
26	1394.508292	1403.713072	1524.152584	1534.750737
27	2566.231309	2568.204887	2611.877555	2628.310468



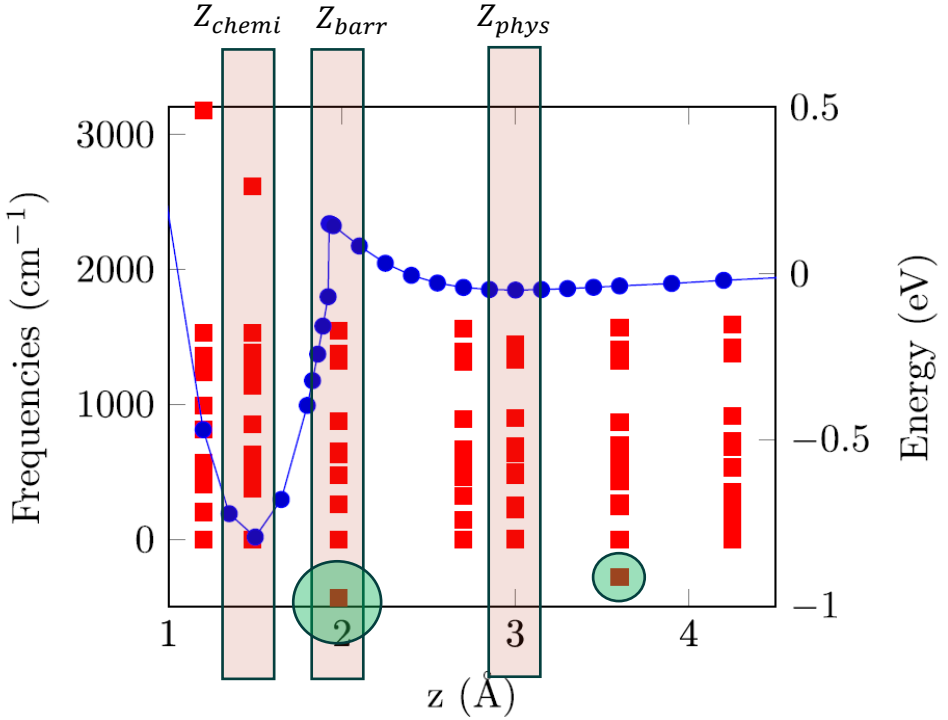


Fig. 2. In red (left y-axis) phonon frequencies at gamma point in the Irreducible Brillouin Zone as a function of the distance of H atom to the surface using vdW-RVV10 functional after application of Acoustic Sum Rule. In blue (right y-axis), the interaction potential with vdW-RVV10 functional as a function of the distance of the H atom to the surface. The barrier is at 2 Å, the chemisorption well is at 1.5 Å and the physisorption well is at 3 Å.

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Mixed Quantum Classical Method

Closely Coupled Wave Packet (CCWP) Dynamics

- $g_i(z)$ are H atom and phonon couplings on a grid
- Here, Q_i and P_i describes position and momenta of vibrational mode ω_i; treated classically.
- V_0 static potential
- H atom is defined by a Gaussian
- Varied incident energy and temperature via equipartition theorem.

$$\hat{H} = \hat{T} + V_{tot} + \hat{H}_{bath}$$

$$\hat{H}_{bath} = \sum_i \frac{1}{2}\omega_i(z)(Q_i^2 + P_i^2)$$

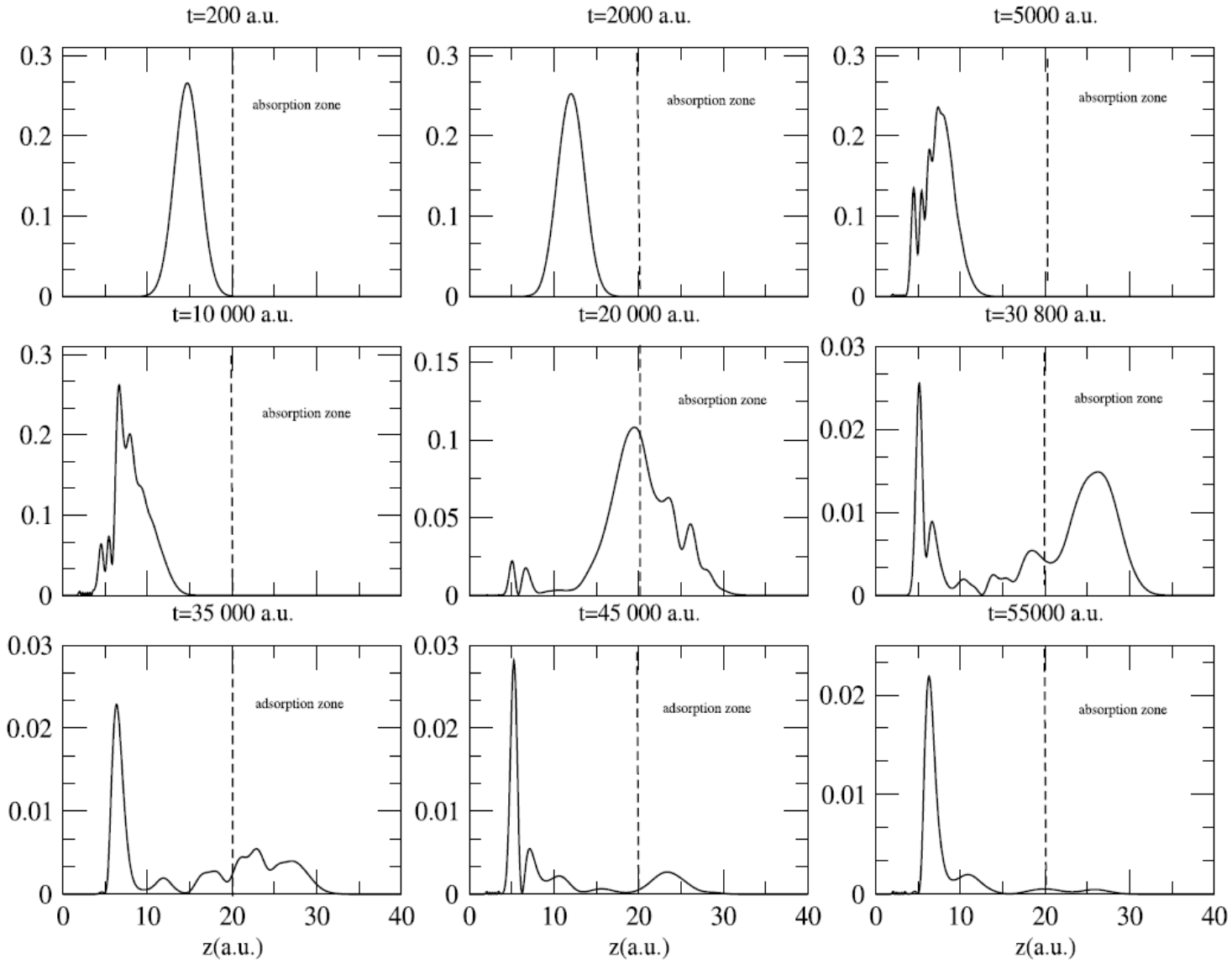
$$V_{tot}(z) = V_0(z) + \sum_i g_i(z)\, Q_i$$

$$\dot{Q}_i = \frac{\partial H_{eff}}{\partial P_i} \quad \& \quad \dot{P} = -\frac{\partial H_{eff}}{\partial Q_i}$$

$$H_{eff} = \langle\psi|H|\psi\rangle$$

S. Morisset, Y. Ferro, A. Allouche, J. Chem. Phys. 133 (2010) 044508.

Sticking Dynamics



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Sticking Probabilities

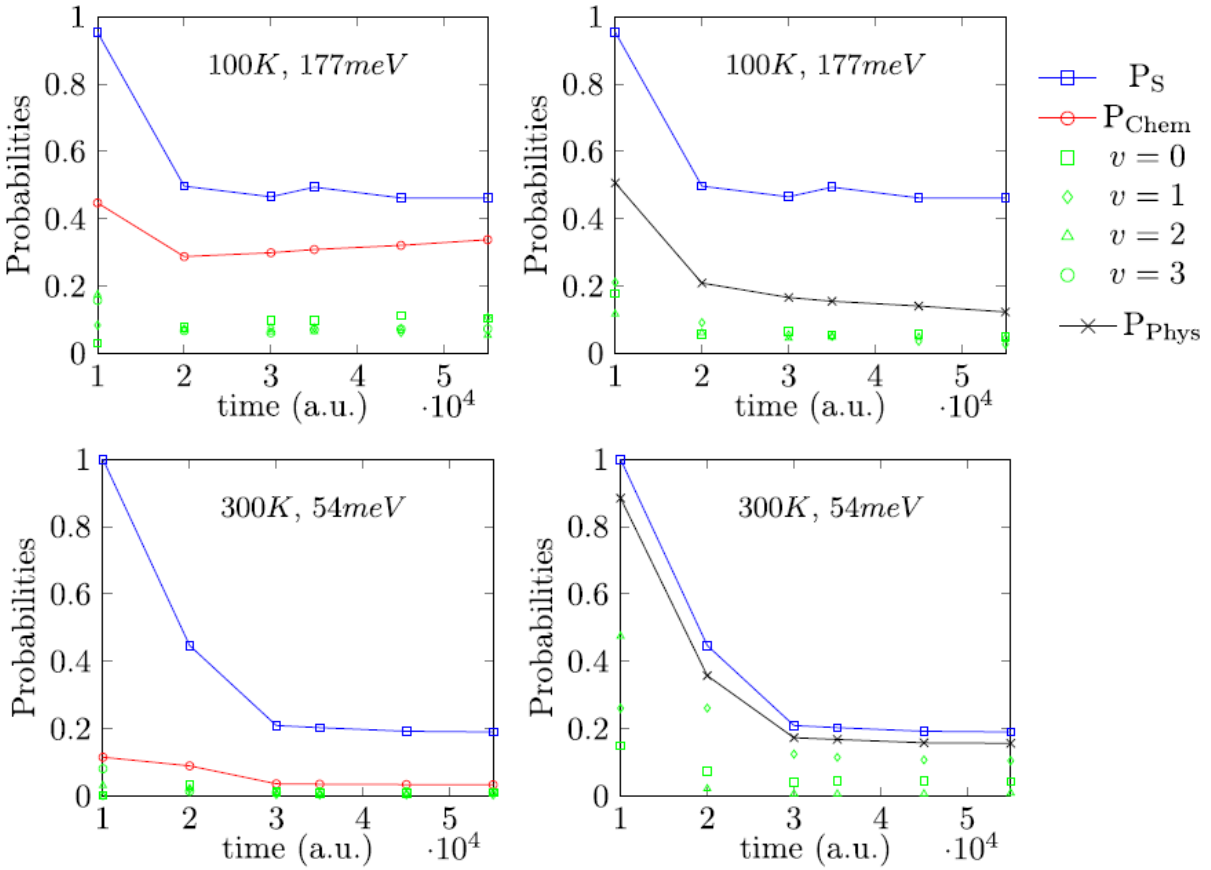
$$P = \frac{\hbar}{m}\int_0^{\infty} Im\left[\Psi^*(z_{abs},t)\left(\frac{\partial\Psi(z,t)}{\partial z}\right)_{z=z_{abs}}\right]dt \qquad P_S = 1 - P$$

Table 3
Sticking probabilities as a function of surface temperature T and of the initial energy E_0 of the wave packet, $N_{traj} = 400$.

E_0 (meV)	50			177		
Temp (K)	10	100	300	10	100	300
P_S	1.8500×10^{-2}	0.07770	0.1923	0.333	0.461	0.584
P_{Chem}	0.0026×10^{-2}	0.00521	0.0341	0.216	0.338	0.448
P_{Phys}	1.8470×10^{-2}	0.07250	0.1581	0.117	0.128	0.135

Table 4
Percentage of the sticking probability on each vibrational states in chemisorption and physisorption.

	Chemisorption				Physisorption		
	v = 0	v = 1	v = 2	v = 3	v = 0	v = 1	v = 2
54 meV 10 K	<0.01%	<0.01%	<0.01%	0.13%	5.1%	92.9%	1.5%
54 meV 100 K	2.6%	0.8%	1.6%	1.6%	14.5%	73.6%	5.0%
54 meV 300 K	5.3%	1.6%	5.0%	5.7%	22.3%	55.2%	4.8%
177 meV 10 K	22.8%	12.0%	14.0%	16.2%	13.1%	7.7%	14.3%
177 meV 100 K	22.5%	23.1%	11.7%	15.8%	10.6%	5.9%	10.0%
177 meV 300 K	20.9%	24.7%	15.0%	16.2%	11.7%	4.7%	6.8%



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Conclusion

- First time, in a single dynamics we have shown sticking on H atom being Chemisorbed and Physisorbed using MQCD
- We show long range dispersion is important for observation of Physisorption potential well

Future Work

- H sticking on graphene is a good system to test new theories, benchmark with Smolyak Grid Method for Bohmian Dynamics (Trajectory Quantic) (HybridTRAJ ANR)

Sabine Morisset
Nathalie Rougeau
David Lauvergnat

Thank you all for your attention!!